Fall 2025 EE582 Homework 02 Report
Nodal-Analysis-based Modeling of Electric Circuits

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Contents

Section A (Numerical Oscillations of Discretization Rules)

One way to analyze the properties of various discretization methods (e.g., in terms of their numerical oscillations) is to solve a lossless first-order system (e.g., inductance or capacitance) when the state variable of the component is suddenly changed. The circuit on the right considers a case where a dc voltage is suddenly applied to a capacitor. Because the source voltage is applied like a step function, we know that the current of the capacitor should look like an impulse, analytically. Let us assume that in the simulations all the variables are zero at $t = 0$ and the voltage source takes its value at $t = \Delta t$.

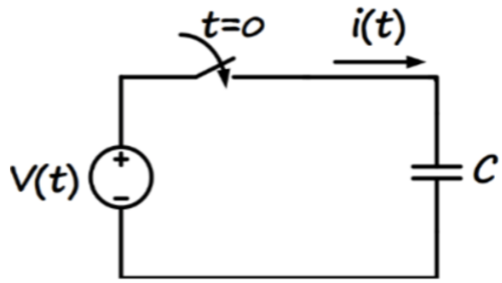

Figure 1: Simple capacitor circuit with DC voltage source

1) Determine the step-by-step solution for the current in the capacitor (in terms of capacitance C, time-step size Δt, and source voltage) using the following discretization rules: a) Forward Euler, b) Backward Euler, c) Trapezoidal.

Trapezoidal Discretization of Capacitor

The final equations for $v(t)$ and $i(t)$ are as follows:

$$\boxed{v(t) = e_h + R_C i(t)} \tag{1}$$

$$\boxed{i(t) = \frac{v(t)}{R_C} - i_h} \tag{2}$$

where:

- $R_C = \frac{\Delta t}{2C}$ is the equivalent resistance.
- $e_h = v(t - \Delta t) + R_C i(t - \Delta t)$ is the history voltage.
- $i_h = \frac{v(t-\Delta t)}{R_C} - i(t - \Delta t)$ is the history current.

Section A: Question 2

Comment on how the simulation time-step size impacts the numerical solution, and which method is preferred in this case.

Section B: Question 1

The parameters of the circuit below are: $R = 10\Omega, L = 20mH, e(t) = 10Vdc$, and the switch is closed at $t = 0$. The circuit is to be solved from $t = 0$ to $t = 10ms$ using:

a) PSCAD software,

b) your own MATLAB code based on the nodal analysis method.

I Appendix: Derivation of Trapezoidal Discretization for Capacitor

To derive the equations $v_{t,\text{cap, trapz}}$ and $i_{t,\text{cap, trapz}}$, the following steps are involved:

- Start with the governing equation for a capacitor:

$$i(t) = C\frac{dv(t)}{dt} \tag{3}$$

- Apply the trapezoidal rule to discretize the relationship between voltage and current:

$$v(t) - v(t - \Delta t) = \frac{\Delta t}{2C}\left[i(t) + i(t - \Delta t)\right] \tag{4}$$

- Define the equivalent resistance R_C and history terms:

$$R_C = \frac{\Delta t}{2C} \tag{5}$$

- Derive the Thevenin (series) and Norton (parallel) companion models.

The final equations for the capacitor are:

$$\boxed{v(t) = e_h + R_C i(t)} \tag{6}$$

$$\boxed{i(t) = \frac{v(t)}{R_C} - i_h} \tag{7}$$

where:

- $e_h = v(t - \Delta t) + R_C i(t - \Delta t)$
- $i_h = \frac{v(t-\Delta t)}{R_C} - i(t - \Delta t)$